

- 6 (a)**
- α : protons decrease by 2, neutron decrease by 2 [B1]
 - β : protons increase by 1, neutron decrease by 1 [B1]
 - γ : no change to number of protons and neutrons [B1]



(b) γ has no charge, so does not experience magnetic force [B1]. So its path should be undeflected [B1]

α and β have opposite charges, so should experience magnetic forces in opposite directions [B1]. So their paths should curve in opposite directions [B1]

(c) (i) Mass is converted [B1] into kinetic energies of nickel-60 nucleus and β particle (also energy of neutrino) as well as electromagnetic energy (photon energy) of the γ radiation [B1]

(ii) Energy released
= (mass loss) c^2
= $(59.9338 - 59.9308 - 5.4348 \times 10^{-4}) (1.66 \times 10^{-27}) (3.00 \times 10^8)^2$ [C1]
= 3.67×10^{-13} J [C1]
= 2.29 MeV [A1]